A Status Note on the Babaioff–Feige ρ-Domination Problem for Self-Maximizing Shares

Verification and partial progress note

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Abstract

This note records a rigorous verification of a proposed approach to the open problem of Babaioff and Feige on feasible, self-maximizing, polynomial-time computable shares that multiplicatively dominate the maximin share for additive valuations. The main status conclusion is negative: the arguments below do *not* solve the polynomial-time open problem. They give partial progress only in the form of clarifications and necessary conditions. In particular, without the polynomial-time requirement the best possible domination factor is exactly the ordinary MMS-feasibility threshold. This shows that any proof of a sharper upper bound for the polynomial-time problem must use computational information. I also prove a small "incentive-closure" necessary condition and explain why that condition, by itself, cannot close the gap left in the Babaioff–Feige paper.

Status of the note. This is not a solution of the Babaioff–Feige problem. The note proves a non-computational reduction, gives a clean endpoint statement for two agents, and isolates a valid but limited necessary condition. The remaining polynomial-time problem remains open in the sense stated in Theorem 9.1.

1 The problem and the MMS-domination formulation

Let M be a finite set of indivisible goods and let $n \geq 2$. A valuation $v : 2^M \to \mathbb{R}_{\geq 0}$ is additive if

$$v(S) = \sum_{g \in S} v(\{g\}) \qquad (S \subseteq M).$$

Write $\mathcal{V}_{\text{add}}(M)$ for the additive valuations on M. Let $\Pi_n(M)$ denote the set of ordered n-partitions of M, allowing empty parts. The maximin share is

$$\text{MMS}_n(v) = \max_{(B_1,\ldots,B_n) \in \Pi_n(M)} \min_{j \in [n]} v(B_j).$$

A share is a family of functions $s = s_{M,n}$ with $s(v, n) \in \mathbb{R}_{\geq 0}$ for each valuation v on M. Babaioff and Feige also impose realizability and item-name independence. These two formal requirements are important for the general theory, but the elementary arguments below use only feasibility, self-maximization, and domination.

Definition 1.1 (Feasibility)**.** *A share s is feasible for additive valuations and n agents if, for every profile $v_1, \ldots, v_n \in \mathcal{V}_{\text{add}}(M)$, there exists a partition $(A_1, \ldots, A_n) \in \Pi_n(M)$ such that*

$$v_i(A_i) \geq s(v_i, n) \qquad \textit{for every } i \in [n].$$

For a share s and a report w, define the acceptable bundles

$$\mathrm{Acc}_s(w,n) = \{S \subseteq M : w(S) \geq s(w,n)\}.$$

If the agent's true valuation is v but she reports w, her worst-case true guarantee under the reported contract is

$$\widehat{s}_v(w,n) = \min_{S \in \mathrm{Acc}_s(w,n)} v(S).$$

When the report is truthful, write $\widehat{s}(v,n) = \widehat{s}_v(v,n)$.

Definition 1.2 (Self-maximization)**.** *A share s is self-maximizing if for every finite M, every $n \geq 2$, and every pair $v, w \in \mathcal{V}_{\mathrm{add}}(M)$,*

$$\widehat{s}(v,n) \geq \widehat{s}_v(w,n).$$

Thus truth-telling maximizes the agent's worst acceptable bundle value.

Definition 1.3 (MMS domination)**.** *For $0 \leq \rho \leq 1$, a share s is ρ-MMS-dominating if*

$$s(v,n) \geq \rho\, \mathrm{MMS}_n(v) \qquad \text{for all finite } M \text{ and all } v \in \mathcal{V}_{\mathrm{add}}(M).$$

Babaioff and Feige phrase their domination question in terms of dominating feasible shares. The MMS-domination formulation above is the numerical form relevant here. The connection is as follows.

Lemma 1.4 (MMS pointwise upper-bounds feasible shares)**.** *If t is feasible for additive valuations and n agents, then*

$$t(v,n) \leq \mathrm{MMS}_n(v) \qquad \text{for every } v \in \mathcal{V}_{\mathrm{add}}(M).$$

Consequently, every ρ-MMS-dominating share s ρ-dominates every feasible share t in the sense that $s(v,n) \geq \rho t(v,n)$ for all v.

Proof. Apply feasibility of t to the identical profile $(v,\ldots,v)$. There is a partition $(A_1,\ldots,A_n)$ with $v(A_i) \geq t(v,n)$ for every i. Therefore

$$\mathrm{MMS}_n(v) \geq \min_i v(A_i) \geq t(v,n).$$

The final statement follows immediately. □

Conversely, Babaioff and Feige's construction used in their nonexistence theorem for a dominant self-maximizing share shows that, for additive valuations, pointwise MMS domination is also forced when one asks to dominate all feasible self-maximizing shares: for each additive valuation v, their family contains a feasible self-maximizing share whose value at v equals $\mathrm{MMS}_n(v)$. Hence a share that ρ-dominates all those shares must satisfy $s(v,n) \geq \rho\, \mathrm{MMS}_n(v)$ for each v. Thus the MMS-domination formulation captures the same numerical threshold for the Babaioff–Feige question.

2 Three thresholds

For fixed n, define the ordinary MMS-feasibility threshold

$$\beta_n = \sup\{\alpha \in [0,1] : \alpha\, \mathrm{MMS}_n \text{ is feasible for additive valuations}\}.$$

Define the non-computational self-maximizing threshold

$$\rho_n^{\mathrm{SM}} = \sup\{\rho \in [0,1] : \text{there exists a feasible, self-maximizing, } \rho\text{-MMS-dominating share}\},$$

and the fixed-n polynomial-time threshold

$$\rho_n^{\mathrm{SM,poly}} = \sup\{\rho \in [0,1] : \text{there exists such a share that is computable in polynomial time}\}.$$

There is also a genuinely uniform version in which one asks for a single polynomial-time computable family working for all n. Denote its threshold by $\rho_{\mathrm{unif}}^{\mathrm{SM,poly}}$. One always has

$$\rho_{\mathrm{unif}}^{\mathrm{SM,poly}} \leq \inf_{n\geq 2} \rho_n^{\mathrm{SM,poly}},$$

and equality is not automatic unless uniformity of the algorithms is separately established.

3 The non-computational problem is exactly ordinary MMS feasibility

The following observation is the central clarification. It is not a solution of the polynomial-time open problem; it explains why the computational condition is the real obstruction.

Lemma 3.1 (Feasibility upper bound)**.** *For every $n \geq 2$,*

$$\rho_n^{\mathrm{SM}} \leq \beta_n \qquad \textit{and} \qquad \rho_n^{\mathrm{SM,poly}} \leq \beta_n.$$

Proof. Suppose s is feasible and ρ-MMS-dominating. For every profile $v_1, \ldots, v_n$, feasibility gives an allocation $(A_1, \ldots, A_n)$ with

$$v_i(A_i) \geq s(v_i, n) \geq \rho\, \mathrm{MMS}_n(v_i) \qquad (i \in [n]).$$

Hence $\rho\, \mathrm{MMS}_n$ is feasible. By definition of β_n, this implies $\rho \leq \beta_n$. The same argument applies with or without polynomial-time computability. □

Theorem 3.2 (Exact non-computational threshold)**.** *For every fixed $n \geq 2$,*

$$\rho_n^{\mathrm{SM}} = \beta_n.$$

Proof. The inequality $\rho_n^{\mathrm{SM}} \leq \beta_n$ is Theorem 3.1. Conversely, let $\alpha < \beta_n$. Then the share $v \mapsto \alpha\, \mathrm{MMS}_n(v)$ is feasible. Babaioff and Feige prove that every feasible share is dominated by some feasible self-maximizing share. Applying their transformation to $\alpha\, \mathrm{MMS}_n$ gives a feasible self-maximizing share s_α with

$$s_\alpha(v, n) \geq \alpha\, \mathrm{MMS}_n(v) \qquad \text{for every additive } v.$$

Thus $\rho_n^{\mathrm{SM}} \geq \alpha$. Taking the supremum over all $\alpha < \beta_n$ gives $\rho_n^{\mathrm{SM}} \geq \beta_n$. □

Corollary 3.3 (What a true upper bound must use)**.** *Any impossibility proof that uses only feasibility, self-maximization, and ρ-MMS domination cannot prove an upper bound below β_n. In particular, an upper bound below the ordinary MMS-feasibility threshold must exploit polynomial-time computability or another genuinely computational restriction.*

Proof. For every $\rho < \beta_n$, Theorem 3.2 gives a feasible self-maximizing ρ-MMS-dominating share, although not necessarily a polynomial-time computable one. Thus no non-computational contradiction is possible below β_n. □

4 Known numerical consequences for the polynomial-time problem

Babaioff and Feige construct, for additive valuations and n agents, a feasible, self-maximizing, polynomial-time computable share that is

$$\frac{2n}{3n-1}\text{-MMS-dominating.}$$

The construction is uniform in n. Hence

$$\rho_{\text{unif}}^{\text{SM,poly}} \geq \inf_{n\geq 2} \frac{2n}{3n-1} = \frac{2}{3}.$$

They also obtain stronger guarantees for small n: a $4/5$ factor for at most four agents and, for two agents, a polynomial-time approximation scheme giving $1-\varepsilon$ for every fixed $\varepsilon > 0$.

For the upper bound, Feige, Sapir, and Tauber give a three-agent additive instance in which no allocation gives all agents more than a $39/40$ fraction of their MMS. Therefore any uniform share that is feasible and ρ-MMS-dominating for all n must satisfy

$$\rho \leq \frac{39}{40}.$$

Thus the currently certified uniform interval is

$$\boxed{\frac{2}{3} \leq \rho_{\text{unif}}^{\text{SM,poly}} \leq \frac{39}{40}}.$$

This interval is a status statement, not a solution.

5 The two-agent endpoint

The fixed two-agent case illustrates the difference between a supremum and an attained maximum.

Proposition 5.1 (Two agents). *For $n = 2$,*

$$\rho_2^{\text{SM,poly}} = 1$$

as a supremum. Assuming $\text{P} \neq \text{NP}$*, no exact polynomial-time computable feasible share attains* $\rho = 1$.

Proof. Babaioff and Feige's two-agent PTAS gives feasible, self-maximizing, polynomial-time computable shares with ratio $1-\varepsilon$ for every fixed $\varepsilon > 0$. Hence the supremum is at least 1, and it is clearly at most 1 by Theorem 3.1, since $\beta_2 = 1$.

Now suppose that a polynomial-time computable feasible share s attains $\rho = 1$. Fix an additive valuation v. Feasibility on the identical profile (v, v) gives a two-partition (A, B) such that

$$v(A) \geq s(v, 2), \qquad v(B) \geq s(v, 2).$$

Therefore $s(v, 2) \leq \text{MMS}_2(v)$. But 1-MMS domination gives $s(v, 2) \geq \text{MMS}_2(v)$. Hence

$$s(v, 2) = \text{MMS}_2(v) \qquad \text{for every additive } v.$$

Exact computation of such an s would exactly compute the two-way partition optimum. Given integer item sizes $a_1, \ldots, a_m$ with total $2B$, one has $\text{MMS}_2(v) = B$ if and only if the items can be partitioned into two bundles of value B. Thus exact computation of MMS_2 solves PARTITION. Unless $\text{P} = \text{NP}$, no exact polynomial-time share attaining $\rho = 1$ exists. □

6 A necessary incentive-closure obstruction

The next construction is a valid necessary condition for any self-maximizing ρ-MMS-dominating share. Its limitation is explained in Section 7.

Definition 6.1 (Incentive closure)**.** *For $v \in \mathcal{V}_{\mathrm{add}}(M)$ and $0 \le \rho \le 1$, define*

$$\Gamma_{\rho,n}(v) = \sup_{w \in \mathcal{V}_{\mathrm{add}}(M)} \min\{v(S) : S \subseteq M,\ w(S) \ge \rho\, \mathrm{MMS}_n(w)\}.$$

Thus $\Gamma_{\rho,n}(v)$ is the largest worst-case true value that valuation v can force by pretending to be some additive report w and using only the promise that the share is at least $\rho\,\mathrm{MMS}_n(w)$.

Proposition 6.2 (Forced guarantee)**.** *Let s be self-maximizing and ρ-MMS-dominating. Then*

$$\widehat{s}(v, n) \ge \Gamma_{\rho,n}(v) \qquad \textit{for every additive } v.$$

If s is also feasible, then for every additive profile $v_1, \ldots, v_n$ there exists an allocation $(A_1, \ldots, A_n)$ such that

$$v_i(A_i) \ge \Gamma_{\rho,n}(v_i) \qquad \textit{for every } i \in [n].$$

Proof. Fix a true valuation v and an additive report w. Since s is ρ-MMS-dominating,

$$s(w, n) \ge \rho\, \mathrm{MMS}_n(w).$$

Therefore

$$\mathrm{Acc}_s(w, n) = \{S : w(S) \ge s(w, n)\} \subseteq \{S : w(S) \ge \rho\, \mathrm{MMS}_n(w)\}.$$

Taking minima of $v(S)$ over these sets gives

$$\widehat{s}_v(w, n) = \min_{S \in \mathrm{Acc}_s(w,n)} v(S) \ge \min\{v(S) : w(S) \ge \rho\, \mathrm{MMS}_n(w)\}.$$

By self-maximization,

$$\widehat{s}(v, n) \ge \widehat{s}_v(w, n).$$

Combining the last two inequalities and then taking the supremum over w proves the first claim.

For the second claim, let $(A_1, \ldots, A_n)$ be an allocation feasible for s on profile $v_1, \ldots, v_n$. Then $A_i \in \mathrm{Acc}_s(v_i, n)$, so

$$v_i(A_i) \ge \widehat{s}(v_i, n) \ge \Gamma_{\rho,n}(v_i)$$

for every i. □

Lemma 6.3 (Basic bounds)**.** *For every additive valuation v,*

$$\rho\, \mathrm{MMS}_n(v) \le \Gamma_{\rho,n}(v) \le \mathrm{MMS}_n(v).$$

Proof. For the lower bound, use the report $w = v$ in Theorem 6.1. Then every bundle in the inner minimization has v-value at least $\rho\, \mathrm{MMS}_n(v)$.

For the upper bound, fix an arbitrary report w and let $(P_1, \ldots, P_n)$ be an MMS partition for w. Each P_j satisfies

$$w(P_j) \ge \mathrm{MMS}_n(w) \ge \rho\, \mathrm{MMS}_n(w),$$

so every P_j is feasible for the inner minimization defining $\Gamma_{\rho,n}(v)$. On the other hand, at least one P_j has $v(P_j) \le \mathrm{MMS}_n(v)$; otherwise the same partition would certify a value strictly larger than $\mathrm{MMS}_n(v)$ for v. Hence

$$\min\{v(S) : w(S) \ge \rho\, \mathrm{MMS}_n(w)\} \le \mathrm{MMS}_n(v).$$

Taking the supremum over w proves the upper bound. □

7 Why the incentive closure cannot close the polynomial-time gap

The obstruction in Theorem 6.2 is mathematically correct, but it is non-computational. This severely limits what it can prove.

Theorem 7.1 (The closure obstruction equals ordinary MMS feasibility)**.** *Fix $n \geq 2$ and $0 \leq \rho \leq 1$. The guarantee $\Gamma_{\rho,n}$ is feasible for additive valuations if and only if ρ-MMS_n is feasible for additive valuations.*

Proof. If $\Gamma_{\rho,n}$ is feasible, then ρ-MMS_n is feasible by Theorem 6.3, because every allocation giving each agent at least $\Gamma_{\rho,n}(v_i)$ also gives each agent at least $\rho\,\mathrm{MMS}_n(v_i)$.

Conversely, suppose ρ-MMS_n is feasible. By the Babaioff–Feige transformation theorem, there exists a feasible self-maximizing share s that dominates ρ-MMS_n [1]. Applying Theorem 6.2 to this s shows that every additive profile $v_1, \ldots, v_n$ has an allocation $(A_1, \ldots, A_n)$ satisfying

$$v_i(A_i) \geq \Gamma_{\rho,n}(v_i) \qquad (i \in [n]).$$

This is exactly feasibility of $\Gamma_{\rho,n}$. □

Corollary 7.2 (Closure infeasibility cannot separate polynomial time)**.** *Fix $n \geq 2$. If $\rho < \beta_n$, then $\Gamma_{\rho,n}$ is feasible. Consequently, proving infeasibility of $\Gamma_{\rho,n}$ can only reproduce ordinary MMS infeasibility; it cannot by itself prove a polynomial-time upper bound below the ordinary MMS-feasibility threshold.*

Proof. If $\rho < \beta_n$, then ρ-MMS_n is feasible by downward closure of feasibility and the definition of β_n. Apply Theorem 7.1. □

This is the main reason the incentive-closure idea should be viewed as partial progress only. It is useful for organizing necessary conditions, but it cannot resolve the Babaioff–Feige computational question unless supplemented by a genuinely computational argument.

8 A small rigid valuation family

The following example is a useful sanity check for Theorem 6.1. It shows that the closure can force full MMS on some valuations, but it does not yield an impossibility theorem by itself.

Proposition 8.1 (Two-level rigid valuation)**.** *Fix $n \geq 2$ and $1/2 < \rho \leq 1$. Let*

$$M = \{g_1, \ldots, g_{n-1}, a, b\}$$

and define an additive valuation v by

$$v(g_i) = 1 \quad (i = 1, \ldots, n-1), \qquad v(a) = \rho, \qquad v(b) = 1 - \rho.$$

Then

$$\mathrm{MMS}_n(v) = 1 \qquad \textit{and} \qquad \Gamma_{\rho,n}(v) = 1.$$

Proof. The partition

$$\{g_1\}, \ldots, \{g_{n-1}\}, \{a, b\}$$

has every part of v-value 1, so $\mathrm{MMS}_n(v) \geq 1$. Since $v(M) = n$, no n-partition can have all parts strictly larger than 1, so $\mathrm{MMS}_n(v) = 1$.

Now consider the additive report w defined by

$$w(g_i) = 1 \quad (i = 1, \ldots, n-1), \qquad w(a) = w(b) = \frac{1}{2}.$$

The same partition shows $\mathrm{MMS}_n(w) = 1$, and $w(M) = n$ gives equality. If $S \subseteq M$ satisfies $w(S) \geq \rho\,\mathrm{MMS}_n(w) = \rho > 1/2$, then either S contains some g_i, in which case $v(S) \geq 1$, or $S \subseteq \{a, b\}$. In the latter case, since $w(a) = w(b) = 1/2$ and $\rho > 1/2$, the set S must contain both a and b, so $v(S) = 1$.

Thus

$$\min\{v(S) : w(S) \geq \rho\,\mathrm{MMS}_n(w)\} \geq 1,$$

so $\Gamma_{\rho,n}(v) \geq 1$. By Theorem 6.3, $\Gamma_{\rho,n}(v) \leq \mathrm{MMS}_n(v) = 1$. Hence $\Gamma_{\rho,n}(v) = 1$. □

9 Remaining problem

The rigorous conclusions above leave the following problem open.

Question 9.1 (Babaioff–Feige polynomial-time threshold)**.** *Determine the largest uniform ρ for which there exists a share family for additive valuations that is feasible, self-maximizing, polynomial-time computable, and ρ-MMS-dominating. Equivalently, determine whether*

$$\rho_{\mathrm{unif}}^{\mathrm{SM,poly}} = \frac{2}{3},$$

or construct a uniform polynomial-time computable feasible self-maximizing share with domination factor strictly larger than $2/3$*.*

Theorem 3.2 shows that this question is not resolved by the non-computational part of the theory: if the polynomial-time requirement is removed, the answer is β_n for fixed n. Theorems 7.1 and 7.2 show that the incentive-closure obstruction is likewise too weak to prove a computational separation by itself.

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